

Gear Garage

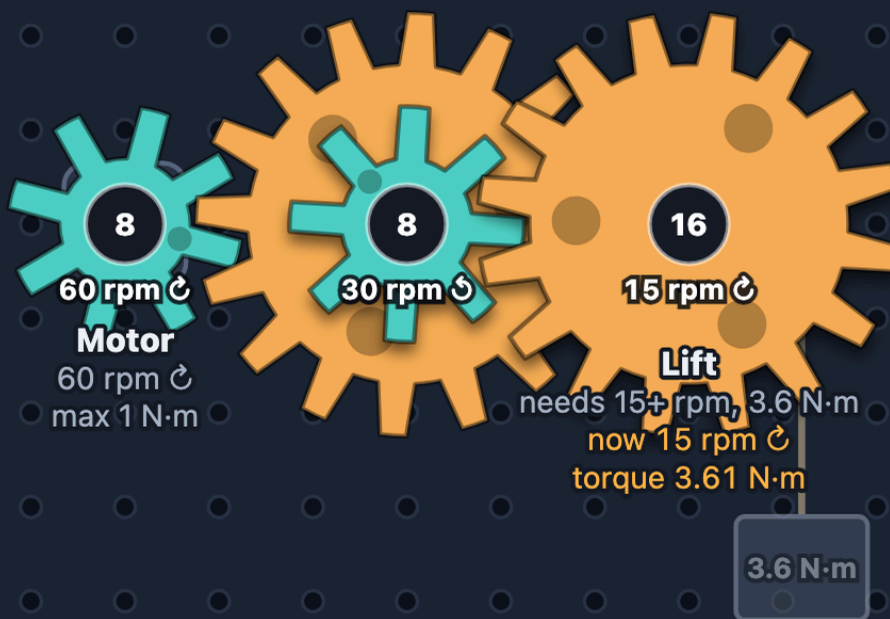
Link gears so a machine turns at the right speed, in the right direction and with enough force: from gear ratios to compound trains and efficiency.

UK Years 8–13

Ages 12–18

US grades 7–12

D&T · Physics



Inside this pack

1. Overview: learning outcomes, curriculum links
2. How the game works (teacher background), key words
3. Level-by-level guide with 3-star solutions
4. Lesson 1: Gears, speed and direction (Years 8–9)
5. Lesson 2: Gear trains, torque and power (Years 10–13)
6. Worksheet 1 and Worksheet 2 (photocopiable)
7. Answers and mark scheme

Play it free

dowplay.com/games/gear-garage

Runs in any browser on Chromebooks, iPads, laptops and phones. No accounts, no ads, nothing to install. Progress is saved only on the device being used.

A separate PDF has the worksheets on their own, ready to print for students.

1 · OVERVIEW

What students learn

Gear Garage turns mechanisms into puzzles. Students place gears on a pegboard to connect a motor to a machine (a wheel, a conveyor, a fan, a winch, a camshaft, a turntable, a garage door, a lift) so that it turns at the right speed, in the right direction and with enough force, within a budget. The gears mesh, jam and stall exactly as real ones would. Its eight levels run from Year 8 to Year 13.

By the end, students can

- explain why meshing gears turn in **opposite directions**, and use **idlers**
- calculate output speeds from the **numbers of teeth**
- explain the trade between **speed and torque**
- show that idlers don't change the overall ratio
- calculate the ratio of a **compound gear train**
- use $v = \omega r$ for a rack and pinion, and $P = T \omega$ with efficiency

You will need

- One device per pair (any browser)
- A projector for the starters
- Worksheet 1 or 2 for each student

Timing

Each lesson plan fills 60 minutes. The levels also work as a homework or a club activity: each takes 5–15 minutes.

Curriculum links

Course	Content	Levels
KS3 Science (England)	Forces: the moment as the turning effect of a force; simple machines give a bigger force at the expense of a smaller movement	3–5
KS3 Design & Technology	How more advanced mechanical systems enable changes in movement and force	All
GCSE Physics / Combined Science	Moments, levers and gears; work done and power; efficiency	3–8
GCSE Design & Technology	Mechanical devices: changing the type of motion and the size and direction of forces; gear trains, idlers, compound gears, rack and pinion	1–7
A-level Physics / Engineering	Rotational motion: torque, angular speed and power ($P = T \omega$); efficiency of transmissions	7–8
US (NGSS)	MS-ETS1-1, MS-ETS1-2, HS-ETS1-3: designing solutions within criteria and constraints, such as cost and size	All

2 · TEACHER BACKGROUND

How the game works

The pegboard follows real gear geometry:

1. An N-tooth gear has a radius of $N \div 8$ holes, so two gears mesh when their axles are $(N_1 + N_2) \div 8$ holes apart: two 16-tooth gears mesh 4 holes apart.
2. Meshing gears turn in opposite directions. Their teeth move at the same speed, so speed $\times$ teeth is the same for both gears.
3. Three gears meshing in a ring **jam**. The motor **stalls** if the machine needs more torque than it can give.

A machine's speed counts as right within 1% either way. Torque (turning force) goes up by the same factor the speed goes down.

The parts

Part	Details
Gears	8, 16, 24, 32 and 40 teeth (radius 1, 2, 3, 4 and 5 holes), costing £2, £3, £4, £5 and £6
Motor	Turns at a set speed (rpm) and direction, and can give up to a set torque (N·m)
Machines	Each needs a speed, a direction or a torque (some need two of these)
Compound gears	Levels 6 and 8: two gears stacked on one axle turn together at the same speed
Rack	Level 7: a straight bar of teeth. Holes are 1 cm apart there, so a gear's radius is $N \div 8$ cm
Efficiency	Level 8: each mesh passes on only 95% of the power

Reading the screen

The gears really turn, with their teeth meshing. Ticking **Show the numbers** labels every gear with its speed (rpm) and direction, and every machine with what it needs, what it's getting and its torque. The worksheet figures are taken straight from that view.

Stars

1 star for a working machine, 2 for staying within the budget, 3 for matching the cheapest design we know. Students sometimes match it with a different set of gears.

What the model leaves out

Real gears lose a little energy at every mesh (the game only models this on level 8), have a little slack (backlash) between teeth, and need stronger, wider teeth for big torques. In the game, gears never wear or break.

Key words

Mesh: two gears' teeth interlocking. **Driver / driven:** the gear that pushes, and the gear being pushed. **Idler:** a gear in between that changes direction but not the overall speed. **Torque:** turning force (N·m). **rpm:** revolutions per minute. **Compound train:** two gears on one axle, so the ratios multiply. **Rack and pinion:** a gear driving a straight bar of teeth. **Efficiency:** power out $\div$ power in.

3 · LEVEL GUIDE

Level by level

"3 stars" is the cheapest design our checker found. Keep it to yourself until students have had a go: the fun is in the search.

1 First Gear

Year 8 · Meshing gears turn each other

- Watch for** Gears that don't quite touch, or that overlap and won't fit: they must be exactly the right distance apart.
- 3 stars, £6** Two gears whose teeth add up to 32 (16 + 16, or 8 + 24), since the axles are 4 holes apart. Three 8-tooth gears also cost £6.
- Ask** "Which way does the wheel turn compared with the motor? Why?"

2 Same Way Round

Year 8 · Idler gears change direction

- Watch for** Two gears (the drum turns the wrong way), or an even number of gears in the chain.
- 3 stars, £8** An 8-tooth gear on the motor, a 24-tooth idler, and an 8-tooth gear on the drum.
- Ask** "What would a fourth gear do?"

3 Need for Speed

Years 8–9 · Speeding up with gears

- Watch for** A big gear on the fan, which slows it down.
- 3 stars, £6** A 24-tooth gear on the motor and an 8-tooth gear on the fan: $20 \text{ rpm} \times 24 \div 8 = 60 \text{ rpm}$.
- Ask** "What does the fan give up to go faster?" (Torque: 1.67 N·m instead of the motor's 5.)

4 Heavy Lifting

Year 9 · Trading speed for torque

- Watch for** Gears that keep the speed the same: the 2 N·m motor stalls against the 5 N·m winch.
- 3 stars, £7** An 8-tooth gear on the motor and a 32-tooth gear on the winch: 4 times slower (7.5 rpm) and 4 times the torque (8 N·m).
- Ask** "What's the smallest slow-down that would work?" (2.5 times.)

5 Round the Engine

Years 9–10 · Idlers don't change the ratio

- Watch for** The camshaft turning the wrong way, or a chain that crashes into the engine block.
- 3 stars, £15** An 8-tooth gear on the crankshaft, then 24, 16 and 16, and a 16-tooth gear on the camshaft: 40 rpm becomes 20 rpm, clockwise.
- Ask** "Why don't the gears in the middle change the camshaft's speed?"

6 Compound Gears

Years 10–11 · Ratios multiply

- Watch for** Trying one pair of gears: the biggest single slow-down is $40 \div 8 = 5$ times.
- 3 stars, £12** 8 teeth driving 24, then an 8-tooth gear stacked on the 24's axle driving a 24-tooth gear on the turntable: $36 \text{ rpm} \div 9 = 4 \text{ rpm}$, with 45 N·m of torque.
- Ask** "How would you slow it down 15 times?" ($8 \rightarrow 24$, then $8 \rightarrow 40$.)

7 Garage Door

Years 11–12 · Rack and pinion, $v = \omega r$

- Watch for** Thinking the gear against the rack decides the speed. In a simple chain, it's the gear on the motor.
- 3 stars, £8** A 24-tooth gear (radius 3 cm) on the motor and a second 24-tooth gear against the rack: $\pi \text{ rad/s} \times 3 \text{ cm} = 9.4 \text{ cm/s}$.
- Ask** "Why doesn't the size of the gear on the rack matter?"

8 Every Mesh Counts

Years 12–13 · Efficiency multiplies

- Watch for** A simple train with idlers: every mesh loses 5%, and the motor stalls.
- 3 stars, £10** A compound train, $8 \rightarrow 16$ then $8 \rightarrow 16$: four times slower in just two meshes, giving 15 rpm and 3.61 N·m.
- Ask** "What would one more mesh do?" (3.43 N·m: the motor stalls.)

A note on "gear ratio". Textbooks usually define the gear ratio as driven teeth $\div$ driver teeth (how many times slower the output turns). The game's hints talk about how many times *faster* the output turns (driver $\div$ driven). This pack avoids the ambiguity by writing **output speed = input speed $\times$ driver teeth $\div$ driven teeth**. Use whichever convention your course uses.

4 · LESSON PLAN · YEARS 8–9 · 60 MINUTES

Lesson 1: Gears, speed and direction

Objectives

- Predict which way each gear in a chain turns.
- Work out a gear's speed from the numbers of teeth.
- Explain how gears trade speed for turning force.

Resources

- Worksheet 1 (one each)
- Devices, one per pair
- Gear Garage on the projector

Time	Activity
Starter 10 min	Open Level 1, First Gear . Ask which way the wheel will turn if one gear on the motor meshes with one on the wheel. Build it and watch: opposite ways. Then Level 2, Same Way Round : "How do we make the drum turn the same way as the motor?" Take ideas before adding an idler.
Main 1 15 min	Worksheet 1, Part A : directions in a chain, speeding up and slowing down, and how far apart gears must be.
Main 2 25 min	Pairs play levels 1–4 and fill in the record table (Part B). Before pressing Check, one partner predicts the output's speed and direction.
Plenary 10 min	Part C together: the Round the Engine figure. Why do the three 16-tooth gears all turn at 20 rpm? Exit ticket: <i>what do you gain and lose when a small gear drives a big one?</i>

Support

Use the in-game hints. Ask students to count along the chain out loud: "clockwise, anticlockwise, clockwise..."

Stretch

Tick Show the numbers and predict every gear's speed before checking. Try Level 5, Round the Engine.

Misconceptions to listen for

- "A bigger gear is always faster." A big gear driving a small one speeds it up; a small gear driving a big one slows it down.
- "Idler gears change the speed." Only the first and last gears decide the overall speed in a simple chain.
- "Faster means stronger." Gears trade one for the other: slower output, more torque.

Homework: Worksheet 1, Part D, and get 3 stars on Level 4.

5 · LESSON PLAN · YEARS 10–13 · 60 MINUTES

Lesson 2: Gear trains, torque and power

Objectives

- Calculate the ratios of simple and compound gear trains.
- Use $v = \omega r$ for a rack and pinion.
- Use $P = T \omega$ and efficiency to explain why fewer meshes are better.

Resources

- Worksheet 2 (one each)
- Devices, one per pair
- Gear Garage on the projector

Time	Activity
Starter 10 min	Show Level 6, Compound Gears : "The turntable must turn 9 times slower, but the biggest pair only gives 5 times. What can we do?" Show a gear stacked on another's axle, and how the two slow-downs multiply.
Main 1 15 min	Worksheet 2, questions 1–2 : idlers and compound trains, using the game's numbers.
Main 2 25 min	Pairs play levels 5–8 . Use questions 3–4 to guide Garage Door and Every Mesh Counts: calculate first, then build.
Plenary 10 min	Plan question 5 together: designing a gearbox. Finish it as homework.

Support

Give students a table to fill in: gear, teeth, speed, direction. In Garage Door, convert rpm to rad/s together first.

Stretch

Find a design for Every Mesh Counts that isn't compound, and explain why it stalls. Research how a bicycle's gears or a clock's gear train work.

Misconceptions to listen for

- "Gears create energy." They trade speed for torque; power out is never more than power in.
- "The gear on the rack sets the door speed." In a simple chain every gear's teeth move at the same speed.
- "rpm can go straight into $v = \omega r$." ω must be in radians per second: $\text{rpm} \times 2\pi \div 60$.

Homework: Worksheet 2, question 5 (6 marks).

Gears, speed and direction

Name: _____

Class: _____

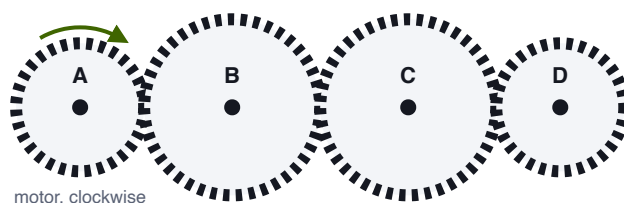
Date: _____

Meshing gears turn in **opposite directions**. $\text{output speed} = \text{input speed} \times \text{driver teeth} \div \text{driven teeth}$

A small gear driving a big gear turns it **slower** but with more **torque** (turning force).

Part A · Gears

A1 Gear A is on a motor that turns clockwise.



a) Which way does each of gears B, C and D turn? [2]

b) How could you make gear D turn clockwise? [1]

A2 A 24-tooth gear on a motor turning at 20 rpm drives an 8-tooth gear.

a) How many times faster does the 8-tooth gear turn? [1]

b) What is its speed in rpm? [1]

A3 An 8-tooth gear on a motor turning at 30 rpm drives a 32-tooth gear on a winch.

a) What is the winch's speed? [1]

b) The motor gives 2 N·m of torque. Ignoring friction, what torque does the winch get? [2]

A4 In Gear Garage, two gears mesh when their axles are $(\text{teeth of one} + \text{teeth of the other}) \div 8$ holes apart. How far apart must these be?

a) two 16-tooth gears b) an 8-tooth and a 24-tooth gear c) an 8-tooth and a 32-tooth gear [3]

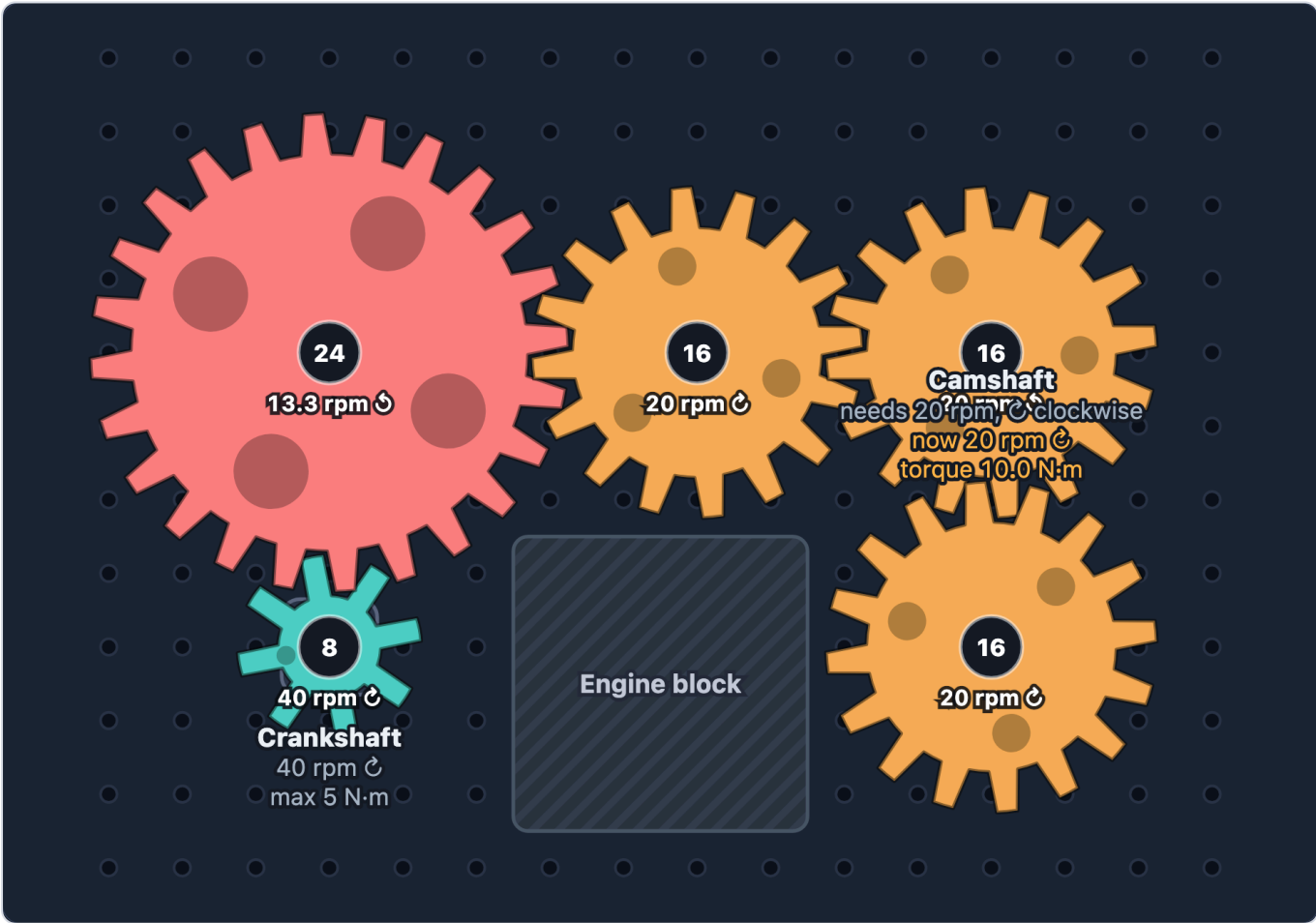
Part B • Play levels 1 to 4

Before you press **Check my gears**, predict which way and how fast the machine will turn. Record your best design for each level.

Level	Gears I used	Cost	Stars	What went wrong, and how I fixed it
1		£		
2		£		
3		£		
4		£		

Part C • Read the numbers

This is **Round the Engine**, solved, with **Show the numbers** ticked. The crankshaft's 8-tooth gear turns at 40 rpm and drives a chain of gears around the engine block to the camshaft. Each gear shows its number of teeth and its speed.



C1 The 24-tooth gear turns at 13.3 rpm. Show how this is worked out from the crankshaft's speed. [2]

C2 The three 16-tooth gears all turn at 20 rpm. Why do they all turn at the same speed? [1]

C3 The camshaft turns at half the crankshaft's speed. Explain this using just the first and last gears. [2]

C4 There are five gears in the chain. Does the camshaft turn the same way as the crankshaft? How do you know? [2]

Part D • Explain

D1 A bicycle has gears. Explain why a low gear makes it easier to ride up a steep hill, and what the rider gives up. [3]

Gear trains, torque and power

Name: _____

Class: _____

Date: _____

Output speed = input speed $\times$ driver teeth $\div$ driven teeth · torque goes up by the factor the speed goes down
 ω (rad/s) = rpm $\times 2\pi \div 60$ · $v = \omega \times r$ · $P = T \times \omega$ · efficiency = power out $\div$ power in
 In Gear Garage, an N-tooth gear has a radius of $N \div 8$ cm on the Garage Door level.

1 Idler gears (Round the Engine). An 8-tooth gear on a 40 rpm crankshaft drives, in turn, gears with 24, 16, 16 and 16 teeth. The last 16-tooth gear is on the camshaft.

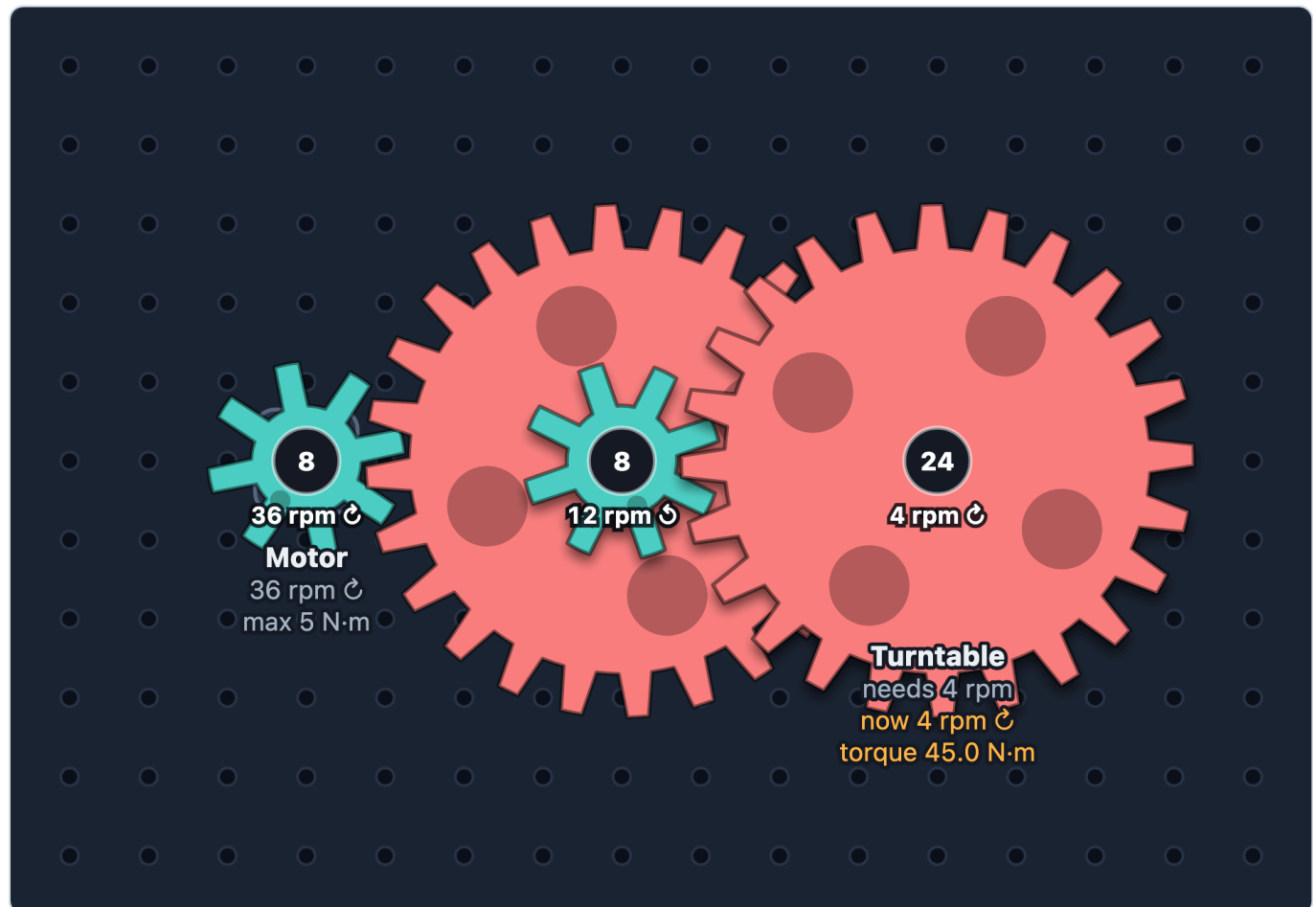
a) What is the camshaft's speed?

[1]

b) Show that the 24-tooth gear has no effect on the camshaft's speed.

[2]

2 Compound gears. This is the solved Compound Gears level. The 24-tooth gear in the middle has an 8-tooth gear stacked on the same axle.



a) Why does the stacked 8-tooth gear turn at 12 rpm? [1]

b) How many times slower than the motor does the turntable turn? Show that the two stages multiply. [2]

c) The motor gives 5 N·m. Ignoring friction, what torque does the turntable get? [1]

d) Using 8-, 24- and 40-tooth gears, design a compound train that turns 15 times slower. [2]

3 Rack and pinion (Garage Door). The motor turns at 30 rpm. The door must rise at 9–10 cm/s.

a) Convert 30 rpm to rad/s. [1]

b) A 24-tooth gear on the motor has a radius of 3 cm. How fast do its teeth move? [1]

c) Which of the 8-, 16-, 24- and 32-tooth gears on the motor gives a door speed of 9–10 cm/s? Show your working. [2]

d) In a simple chain, why doesn't the size of the gear touching the rack change the door's speed? [2]

4 Efficiency (Every Mesh Counts). The motor gives 1 N·m at 60 rpm. The lift needs 3.6 N·m at 15 rpm or faster. Each mesh passes on 95% of the power.

a) What is the motor's power output? [2]

b) How many times slower must the lift turn? What torque would it get with no losses? [2]

c) A compound train does this in two meshes. What torque does the lift get? [2]

d) What would a third mesh do? [2]

5 Evaluate. A machine needs a 3000 rpm motor turned down to 100 rpm. Evaluate three ways of doing it: a single pair of gears, a simple train with idlers, and a compound train. Consider size, number of meshes, efficiency and direction, and use evidence from Gear Garage. [6]

7 · ANSWERS

Worksheet 1 answers

Part A

A1 a) B **anticlockwise**, C **clockwise**, D **anticlockwise** (2 marks for all three).
b) Add one more gear (an idler) anywhere in the chain, or remove one, so there's an odd number of gears.

A2 a) $24 \div 8 = 3$ times.

b) $20 \times 3 = 60$ rpm.

A3 a) $30 \times 8 \div 32 = 7.5$ rpm.

b) Four times slower, so four times the torque (1): $2 \times 4 = 8$ N·m (1).

A4 a) $32 \div 8 = 4$ holes. b) $32 \div 8 = 4$ holes. c) $40 \div 8 = 5$ holes.

Part C

C1 $40 \times 8 \div 24$ (1) = **13.3 rpm** (1).

C2 Gears of the same size meshing together turn at the same speed.

C3 Only the first and last gears decide the speed (1): $40 \times 8 \div 16 = 20$ rpm, half the speed (1).

C4 **Yes** (1): each mesh reverses the direction, and with five gears (an odd number) the last turns the same way as the first (1).

Part D

D1 In a low gear the pedals turn more times for each turn of the wheel (1), so the wheel turns slower but with more torque (1), making it easier to push up the hill; the rider gives up speed (1).

More from DowPlay

Free engineering games for ages 12–18: **Light It Up** (circuits), **Bridge Builder** (structures) and **Grid Manager** (energy), each with a free teacher pack. Ideas for new games? Get in touch at dowplay.com/about#contact.

Worksheet 2 answers

1 a) $40 \times 8 \div 16 = 20$ rpm.

b) Going through the 24: $40 \times 8 \div 24 = 13.3$ rpm, then $13.3 \times 24 \div 16 = 20$ rpm (1). The 24s cancel: $40 \times (8 \div 24) \times (24 \div 16) = 40 \times 8 \div 16$ (1).

2 a) It's on the same axle as the 24-tooth gear, so they turn together.

b) $24 \div 8 = 3$ at each stage (1); $3 \times 3 = 9$ times: $36 \rightarrow 12 \rightarrow 4$ rpm (1).

c) $5 \times 9 = 45$ N·m (as the game shows).

d) $8 \rightarrow 24$ (3 times) (1), then a stacked $8 \rightarrow 40$ (5 times): $3 \times 5 = 15$ (1).

3 a) $30 \times 2\pi \div 60 = \pi \approx 3.14$ rad/s.

b) $v = 3.14 \times 3 = 9.4$ cm/s.

c) Radii 1, 2, 3 and 4 cm give 3.1, 6.3, 9.4 and 12.6 cm/s (1); only the **24-tooth gear** works (1).

d) Meshing teeth move at the same speed (1), so in a simple chain every gear's teeth, and the rack, move at the motor gear's tooth speed (1).

4 a) $\omega = 60 \times 2\pi \div 60 = 2\pi$ rad/s (1); $P = 1 \times 2\pi = 6.3$ W (1).

b) $60 \div 15 = 4$ times (1); $1 \times 4 = 4$ N·m (1).

c) $4 \times 0.95 \times 0.95$ (1) = **3.61 N·m**, just enough (1).

d) $4 \times 0.95^3 = 3.43$ N·m (1), less than 3.6 N·m, so the motor stalls (1).

Question 5 mark scheme (6 marks)

Level	Description
3 (5–6 marks)	A detailed evaluation of all three options, comparing size, meshes, efficiency and direction, with calculations or evidence from the game. Reaches a justified conclusion (usually a compound train).
2 (3–4 marks)	Compares at least two options with some evidence. The conclusion is partial or not fully justified.
1 (1–2 marks)	Simple statements about one or two options, with little comparison.
0	Nothing worthy of credit.

Indicative content

Single pair

- A 30 times slow-down needs one gear 30 times bigger, e.g. 8 and 240 teeth
- Very large and heavy
- Only one mesh, so efficient; output turns the opposite way

Simple train with idlers

- Idlers don't change the ratio, so it's no smaller (Round the Engine)
- Each extra mesh wastes energy (Every Mesh Counts)
- Useful only to bridge a gap or change direction

Compound train

- Ratios multiply, e.g. $5 \times 6 = 30$ in two meshes (Compound Gears)
- Small and efficient: $0.95^2 \approx 90\%$
- Two meshes, so the output turns the same way as the motor

Credit references to real examples: car gearboxes, clocks, bicycles, and to the extra torque the output gets (30 times, less losses).